

Comparative Analysis of Machine Learning Algorithms for Spam Email Classification

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Machine Learning – Course Project Report



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1 Problem Statement

Unsolicited and potentially harmful spam messages pose a significant challenge in digital communication. The objective of this project is to develop an efficient machine learning-based spam email detection system capable of automatically classifying incoming messages as spam or ham. By applying natural language processing (NLP) techniques and supervised learning algorithms, the project aims to enhance email filtering accuracy and reduce user exposure to unwanted or malicious content.

The objectives of this project are :

- To design a machine learning model for accurate spam email detection.
- To apply NLP techniques for effective text preprocessing and feature extraction.
- To evaluate model performance using appropriate classification metrics.
- To develop a lightweight application for real-time spam message identification.

2 Proposed Methodology

The proposed system aims to automatically classify email or text messages as *spam* or *ham* (non-spam) using machine learning and natural language processing (NLP) techniques. The overall methodology consists of several well-defined stages as described below.

2.1 Data Collection and Loading

The dataset used for this project is the **Spam Email Dataset (spam.csv)**, which contains labeled email and SMS messages categorized as *spam* or *ham*. The dataset is loaded into the Python environment using the Pandas library for further analysis and preprocessing.

2.2 Data Cleaning

Unnecessary columns and missing entries are removed to retain only the essential attributes: the message content and its label. The data is then checked for duplicate entries and text encoding issues to ensure consistency and reliability.

2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to gain insights into message distributions and word patterns. Statistical and visual analyses are carried out using the Matplotlib and Seaborn libraries to:

- Observe the ratio of spam to ham messages.
- Analyze average word count differences between the two categories.
- Identify patterns in frequent words using word clouds or bar charts.

2.4 Text Preprocessing

Before applying machine learning, the raw text is preprocessed to convert it into a clean and structured format suitable for modeling. The following steps are used:

- **Lowercasing:** Converts all text to lowercase for uniformity.
- **Tokenization:** Splits sentences into individual words (tokens).
- **Stopword and Punctuation Removal:** Removes common, uninformative words and symbols.
- **Stemming:** Reduces words to their root form using Porter Stemmer from NLTK.

This process minimizes noise and ensures better model generalization.

2.5 Feature Extraction

The cleaned text is transformed into numerical vectors using **TF-IDF (Term Frequency–Inverse Document Frequency)**, which captures the importance of words relative to their frequency across messages. This feature matrix serves as the input to all machine learning models.

2.6 Model Building

Multiple supervised machine learning algorithms are implemented and trained to classify messages. Each model is fitted on the training data and evaluated to identify the most accurate and reliable one. The models used include:

- Multinomial Naïve Bayes (MNB)
- Bernoulli Naïve Bayes (BNB)
- Logistic Regression (LR)
- Support Vector Machine (SVM)
- Decision Tree Classifier (DTC)
- Random Forest Classifier (RFC)
- K-Nearest Neighbors (KNN)
- Gradient Boosting Classifier (GBC)

Each classifier's hyperparameters are fine-tuned where necessary, and comparative analysis is performed to select the model yielding the highest precision and recall for spam detection.

2.7 Model Evaluation

All trained models are assessed using standard classification metrics:

- **Accuracy:** Overall correctness of predictions.
- **Precision:** Ability to correctly identify spam without false alarms.
- **Recall:** Ability to detect all actual spam messages.
- **F1-Score:** Harmonic mean of precision and recall.

A **confusion matrix** is also generated to visualize true and false classifications, ensuring a comprehensive evaluation of performance.

3 Dataset Details

The dataset used for this project is the **Spam Email Dataset (spam.csv)**, which contains labeled email and SMS messages categorized as *spam* or *ham* (non-spam). It is a publicly available dataset commonly used for text classification and spam filtering research.

Dataset Summary

Attribute	Description
Total Messages	5,572
Ham (Non-spam) Messages	4,825
Spam Messages	747
Features	Message Text, Label
File Format	CSV (Comma-Separated Values)
Source	Public Spam Collection Dataset

Table 1: Summary of the Spam Email Dataset

Remarks:

The dataset exhibits a slight class imbalance, with a higher number of ham messages compared to spam messages. During preprocessing, unnecessary columns, duplicates, and missing values were removed. The final dataset was divided into **training** and **testing** sets for model evaluation.

4 Comparative Analysis & Results

This section presents the evaluation and comparison of different machine learning models applied to the spam email detection task. The models were trained on TF-IDF features and tested on unseen data to assess their performance.

4.1 Evaluation Metrics

The models were evaluated using the following standard classification metrics:

- **Accuracy:** Overall correctness of the classifier.
- **Precision:** Fraction of correctly identified spam messages among all predicted spam.
- **Recall:** Fraction of actual spam messages correctly identified.
- **F1-Score:** Harmonic mean of precision and recall.

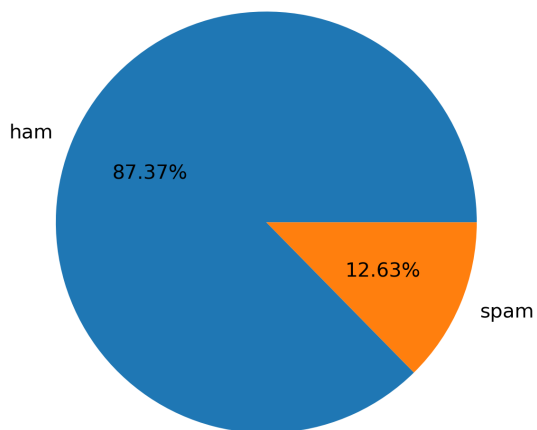
4.2 Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naïve Bayes	96.2%	1.0	0.717	0.835
Bernoulli Naïve Bayes	97.7%	0.991	0.84	0.909
Logistic Regression	95.2%	0.949	0.681	0.793
Support Vector Machine	97.5%	0.974	0.833	0.898
Random Forest Classifier	97%	0.991	0.782	0.874
Decision Tree Classifier	93.4%	0.843	0.623	0.716
K-Nearest Neighbors	90.2%	1.0	0.268	0.423
Gradient Boosting Classifier	95.2%	0.908	0.717	0.801

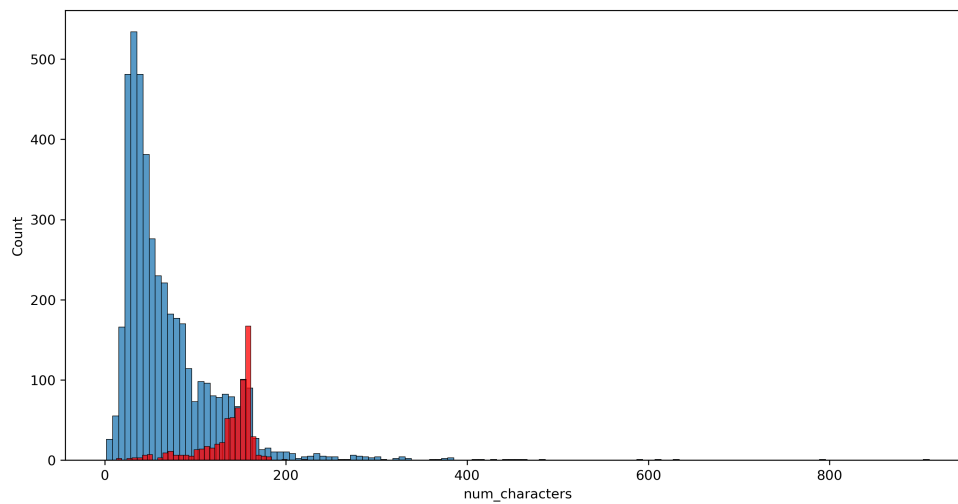
Table 2: Performance Comparison of Machine Learning Models

4.3 Visualization of Results

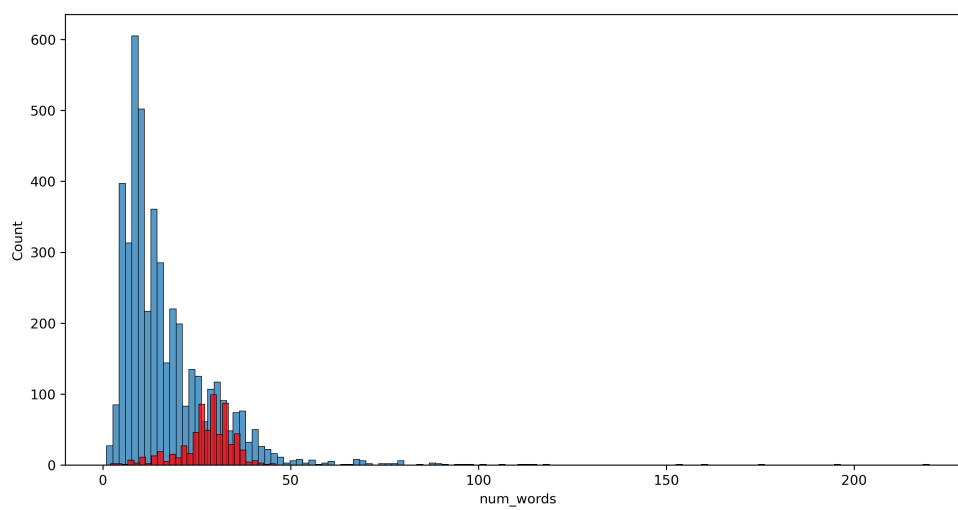
Plot 1:Spam Vs Ham Pie Chart



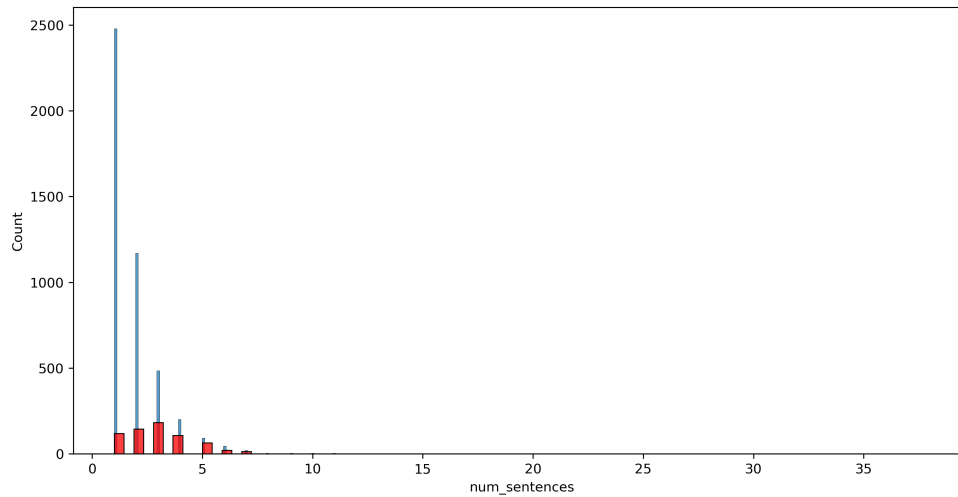
Plot 2:Histogram of Spam characters Vs Ham characters



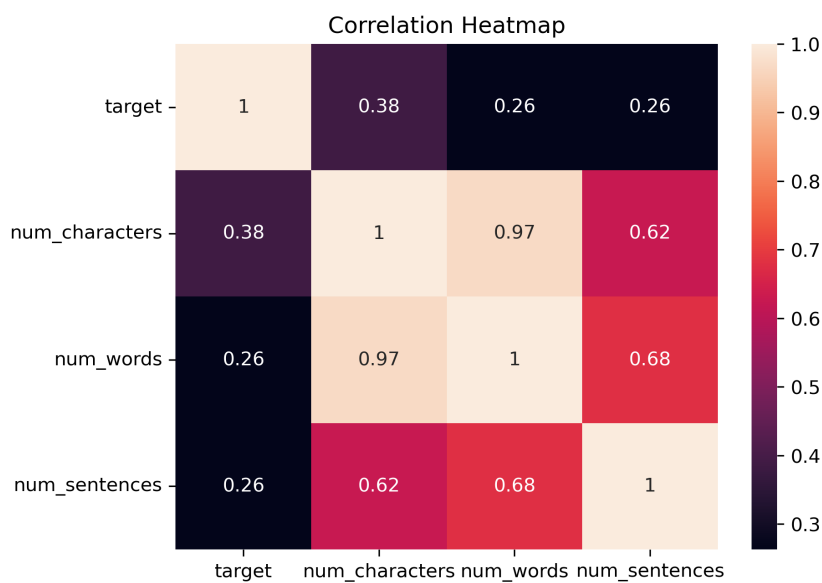
Plot 3:Histogram of Spam words Vs Ham words



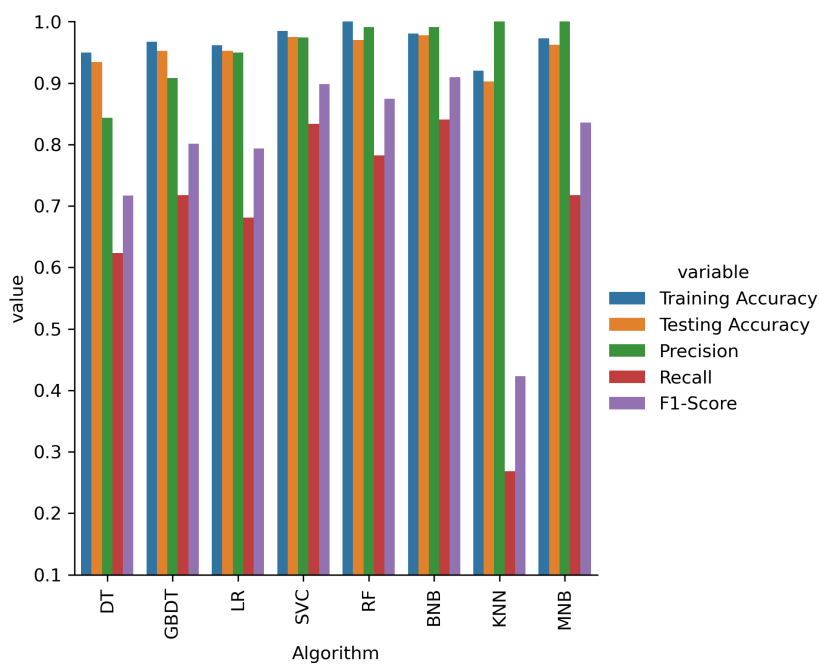
Plot 4:Histogram of Spam sentences Vs Ham sentences



Plot 5:Correlation Heatmap



Plot 6 Accuracy, Recall, Precision and F1_Score Comparison Plot Across Models



4.4 Discussion

From the above results, the **Bernoulli Naive Bayes** model achieved the highest performance, showing excellent balance between precision and recall. **Support Vector Machine (SVM)** and **Random Forest Classifier** also performed competitively. These results demonstrate that NLP-based feature extraction using TF-IDF combined with supervised machine learning algorithms can effectively distinguish spam messages from legitimate ones.

5 Conclusion

This project demonstrates that combining **Natural Language Processing (NLP)** and **Machine Learning (ML)** techniques can effectively classify messages as *spam* or *ham*. Using TF-IDF feature extraction and various supervised models, the **Bernoulli Naive Bayes** achieved the highest accuracy and reliability among all tested algorithms.

The results confirm that text preprocessing and model selection play a vital role in improving spam detection performance.

References

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